

Read Chapters 3 & 6

Unit 2_01_UsefulMethods

Useful Methods for Any Class

An new object copy or clone

- There various ways to create exact copy or clone of an object. I will show some here.
- Assuming that you have a class say Student which has a fully parametrized constructor. Then, s2 is a copy of s1 using:

```
Student s2 = new Student(s1.getId(),s1.getName(),s1.getMajor(),s1.getMinor())
```

Copying an object

- If you already have two objects say s1 and s2 and you want to have that data in s2 to be identical to the data in s1 then this is called copying.
  - In the class student, create a copy instance method that receives the object to copy to.
  - Note: Copying everything, including auto generated data such as id.

```
public copy(Student other){
     other.id=id; other.name=name; other.major=major; other.minor=minor;
}
```

```
s1.copy(s2);
```

Cloning an object

- If you one object, say s1, and you want to create a new object with identical data to the data in s1 then this is called cloning.

```
public Student clone(){
        return new Student (id, name, major, minor);
}
```

```
Student s2 = s1.clone();
```

equals method

- If you have two object, say s1 and s2, and you want to check if they have the same values for all of their attributes:
  - Create the instance method equals that receives an object of Student and compares the values of its attributes with that of the calling objects. True is returned if all are equal otherwise, false is returned.

```
public boolean equals(Student other){
    if (other.id == id and other.name.equals(name) and
            other.major.equals(major) and
            other.minor.equals(minor))
            return true;
    else return false;
}
```

```
System.out.println( “s2 equals s1 is “+ s1.equals(s2));
```

compareTo method

- If you have two object, say s1 and s2, and you want to compare them, say based on the value of the attribute id:
  - Create the instance method compareTo that receives an object of Student and compares the value of its attribute id with that of the calling objects. 0 is returned if they are equal, 1 if the id of s1 is larger, and -1 if the id is smaller.

```
public int compareTo(Student other){
      if (id >other.id)
            return 1;
      else if(id<other.id)
           return -1;
      else return 0;
}
```

```
int result = s1.compareTo(s2);
```